



KONERU LAKSHMAIAH EDUCATION FOUNDATION

Department of Computer Science and Engineering

1ST REVIEW

SMART WASTE MANAGEMENT SYSTEM

A database-driven system that uses simulated real-time waste-level data to monitor bins and manage waste collection.

TEAM

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GUIDED BY

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SYSTEM FLOW



Monitoring



Database



Collection



Analytics



Team & Responsibilities

GROUP 10



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RESPONSIBILITIES

Abstract

Introduction

Existing System



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Problem

Solution

Objectives



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RESPONSIBILITIES

Modules

Software Requirements

Conclusion & Future Scope

SYSTEM OVERVIEW

- A database-driven solution that uses simulated real-time waste-level data to monitor bins.
- Bins crossing a predefined threshold are identified for collection.
- Collection requests are created, assigned and tracked, and history is kept for reports.



Smart Monitoring

Simulated real-time waste levels



Centralized Database

Bin, level and collection data in one place



Collection Tracking

Create, assign and track requests



Historical Records

Waste-level and collection history for reports

WASTE COLLECTION CHALLENGE



Waste generation varies by location and time



Collection needs proper monitoring and timely action



Stored data can improve future collection planning

OUR APPROACH



Smart Bin

01



Waste-Level Data

02



Centralized Database

03



Monitoring

04



Collection Planning

05

Existing System

BEFORE



Fixed Schedules

Collection follows planned schedules



Manual Monitoring

Workers check bins by hand



Unknown Bin Levels

Actual waste level may be unknown before collection



Separate Systems

Basic sensors and route tools, not one system



Delayed Response

Hard to react when a bin fills early



Limited Historical Data

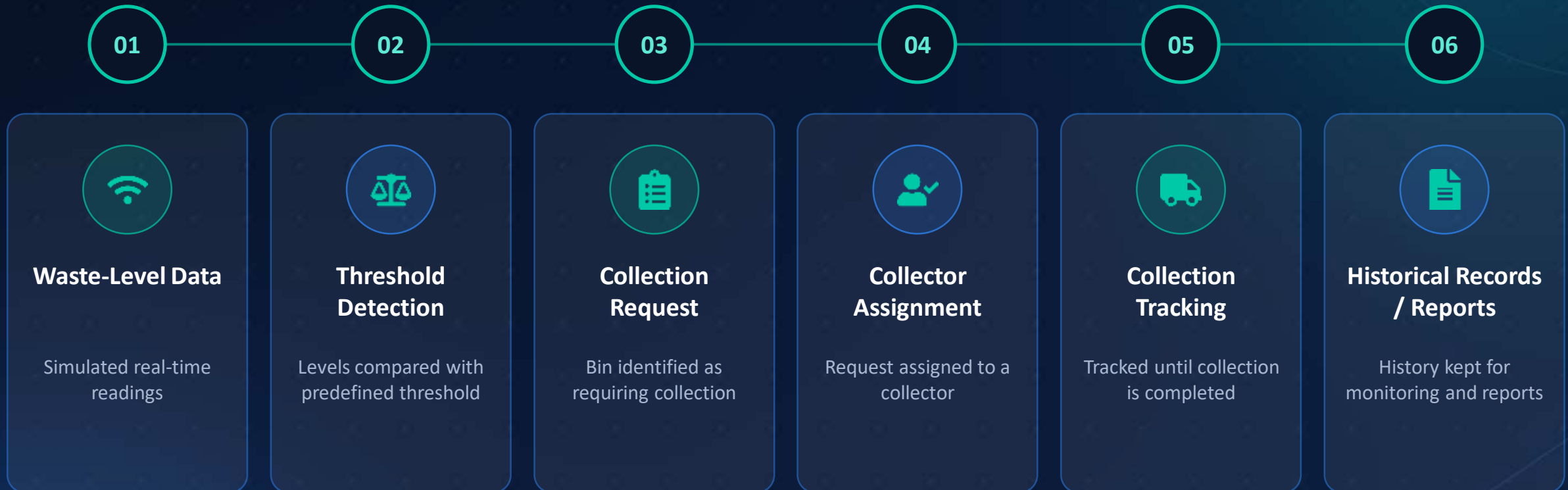
Little organized data for analysis

Problem

THE CORE CHALLENGE



Operational cost increases and the cleanliness of public and residential areas is affected



Bins crossing the predefined threshold are identified, assigned and tracked until collection is completed.

01



Monitor Waste Levels

Use simulated real-time data representing sensor readings

02



Centralize Data

Store waste-bin and collection information in one database

03



Detect Thresholds

Identify bins requiring collection using predefined levels

04



Track Collection

Create, assign and track collection requests

05



Maintain History

Keep waste-level and collection records and generate reports

06



Support Data-Driven Decisions

Rely on actual bin data, not only fixed schedules



User & Role Management

- Registration and login
- User details and roles
- Access control



Smart Bin Management

- Add and manage bins
- Location and capacity
- Status and details



Waste-Level Monitoring

- Store simulated readings
- Track status and thresholds
- Maintain level history



Collection Management

- Create collection requests
- Assign requests to staff
- Track status and history



Collector & Vehicle Management

- Collector details
- Vehicle information
- Assignments and activity records



Reports & Analytics

- Waste-level reports
- Collection statistics
- Bin-wise and location-wise analysis



DATABASE

PostgreSQL

Stores bin, waste-level, collector and collection data



BACKEND

Node.js + Express.js

Develops backend APIs connected to the database



API TESTING

Postman

Tests and verifies the backend REST APIs



DEVELOPMENT

Visual Studio Code

Develops and manages the project code



CONTAINERIZATION

Docker Desktop

Containerizes application and database for consistent deployment



DATA SIMULATION

Node.js

Generates simulated real-time waste-level readings



01

IoT Sensors

Collect actual data from smart bins



02

GPS Tracking

Track collection vehicles



03

Route Optimization

Efficient collection routes



04

Mobile App

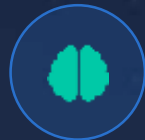
Staff receive and update requests



05

Real-Time Alerts

Notify at critical waste levels



06

Predictive Analysis

Estimate future waste generation



07

Cloud Deployment

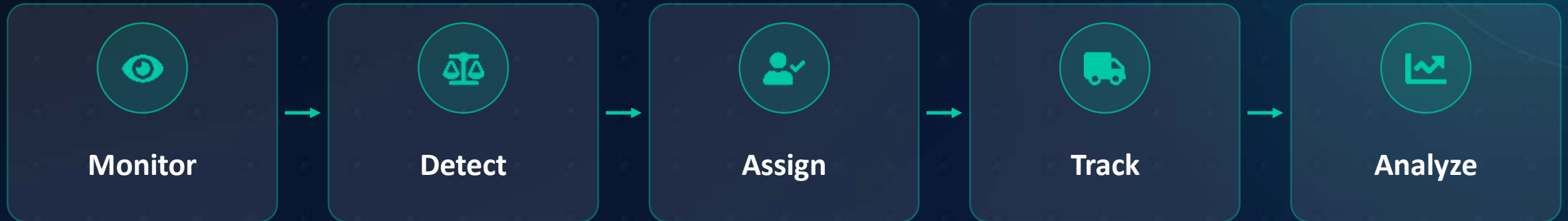
Access from multiple locations



08

Smart City Integration

Larger-scale waste management



SMART WASTE MANAGEMENT SYSTEM

The proposed system provides a structured, data-driven approach to waste collection management. By using simulated real-time waste-level data and a centralized PostgreSQL database, it monitors bin conditions, identifies bins that require collection and maintains collection records. It reduces dependence on fixed schedules and provides organized data that supports better collection planning and decision-making, showing how database technology can make waste management more efficient, organized and reliable.



THANK YOU

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SMART WASTE MANAGEMENT SYSTEM

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